

Safetysquad

Reflection

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1. Introduction

In this document, I present a complete overview of my internship project, my learning process, and my personal development throughout the internship period. The report is divided into several parts to give the reader a clear and structured understanding of both the technical work I conducted and the growth I experienced along the way.

First, I describe the internship context and the goals that were set at the start of the project. This provides the necessary background to understand the choices made during development.

Next, I give a substantive reflection on the project itself, where I discuss what I accomplished, how my work contributes to the client's organization, and what remains to be done.

After that, I include a personal reflection in which I look back on my own learning process. Here, I reflect on the competencies I developed, the challenges I encountered, and the ways in which I grew both professionally and personally.

Finally, the document concludes with a general conclusion that summarizes the most important insights and highlights how this internship has shaped my future ambitions as a developer.

This structure ensures that the reader can easily follow the progression from project context to technical outcomes, to personal development, and to the broader significance of the internship experience.

2. Substantive reflection

Throughout my internship, I contributed to the further development and modernization of the platform by focusing primarily on the backend architecture and the stability of the system. My most significant accomplishment was the redesign and implementation of a new API, built according to principles such as Clean Architecture and Domain-Driven Design. This required a thorough analysis of the existing codebase, identifying structural inconsistencies, and refactoring the database to support clearer domain boundaries and long-term maintainability. These improvements provide the client with a more scalable and future-proof backend, while end users benefit from a more reliable and predictable application experience.

In addition to the architectural redesign, I delivered several concrete features, including Microsoft integrations, deployment preparations, and optimizations across both backend and frontend components. These contributions help the organization streamline internal processes and improve the operational stability of the platform. Importantly, all my contributions were fully deployed to the production environment by the end of my internship, which demonstrates that the work delivered was not only technically sound but also ready for real-world use.

Although the core objectives of my project were completed, there are still opportunities for further refinement. Additional automated testing, extended documentation, and further UI/UX improvements would strengthen the long-term maintainability and usability of the system. These aspects were outside the primary scope of my internship but would be valuable next steps for the organization.

For the future, I would advise the client to continue investing in clear architectural guidelines and consistent technical documentation, especially since the project is regularly handed over to new developers or interns. Establishing a structured onboarding process and maintaining a shared architectural vision will help ensure that the platform evolves in a coherent and sustainable way. Expanding automated testing and monitoring would further enhance reliability and reduce the risk of regressions.

Overall, the project delivered tangible value to the organization and provided a solid technical foundation for future development. The fact that all my work was put into production confirms its practical relevance and impact.

3. Personal reflection

This internship has been an important personal milestone for me. It offered me the opportunity to apply my theoretical knowledge in a real professional environment and to discover how software development functions within a larger organization. Being part of a project with real users and real impact made the experience meaningful and motivating. It also confirmed that I genuinely enjoy working on backend development and architectural challenges, which gives me confidence in the direction of my future career.

Throughout the internship, I developed a wide range of competencies. On a technical level, I strengthened my skills in backend development, API design, and architectural thinking. Working with principles such as Clean Architecture and Domain-Driven Design helped me understand how to structure complex systems in a maintainable way. I also gained hands-on experience with database refactoring, Microsoft integrations, and deployment processes. On a professional level, I improved my communication skills significantly. Presenting my work in both English and Dutch, participating in meetings, and collaborating with different profiles taught me how to communicate clearly and confidently in a team environment.

I also grew personally during this internship. I became more independent in analysing problems, making decisions, and taking responsibility for my work. At the same time, I learned to ask for help when needed and to involve others early when facing uncertainties. This balance between autonomy and collaboration is something I will carry with me into future projects.

Of course, the internship also came with challenges. One of the biggest difficulties was working with an existing codebase created by previous interns. Understanding their choices, navigating unclear structures, and identifying what needed improvement required patience and persistence. I addressed this by taking the time to analyse the code thoroughly, asking targeted questions, and validating my assumptions before implementing changes. Another challenge was presenting my work to different stakeholders. At first, this felt intimidating, but through practice and feedback from my mentor, I became more confident and structured in my communication. These experiences helped me grow not only as a developer but also as a professional.

Overall, this internship has been a valuable and transformative experience. It strengthened my technical abilities, improved my communication skills, and gave me insight into how software is built, maintained, and delivered in a professional environment. It confirmed my interest in backend development and provided a solid foundation for the next steps in my career.